

Taste Shift

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1. Introduction

The goal of the AI-Powered Recipe Website and Mobile App project is to design a complete digital platform for food lovers. The website will be created with the MERN (MongoDB, Express.js, React, and Node.js) stack, whereas the mobile app will be designed with Flutter and Firebase. The main attraction of the platform is an AI-based Chabot that will help users discover recipes, nutritional facts, and cooking procedures. Moreover, the platform will feature categorized recipes, video tutorials, user interaction features, and real-time content updates.

2. Objectives

- Create a feature-rich, user-friendly website with the MERN stack.
- Develop a cross-platform mobile app with Flutter.
- Integrate Firebase for authentication, real-time database, and cloud storage.
- Integrate an AI Chabot to help users with recipe suggestions and cooking instructions.
- Facilitate dynamic content updates, such as recipe videos and user-submitted content.
- Provide seamless user experience on web and mobile platforms.
- Offer interactive engagement elements like reviews, ratings, and comments.

3. Problem Description

Most cooking hobbyists face issues with:

- Looking for appropriate recipes based on dietary requirements and accessible ingredients.
- Relating nutritional contents and health impacts of various meals.
- Adhering to confusing cooking procedures lacking visual support.
- Engaging in a platform offering real-time guidance during cooking.

- Current solutions do not have personalized suggestions, interactive support, and live updates. This project will fill these gaps with an AI-driven Chabot, intelligent search filters, and multimedia content incorporation.

4.Methodology

The project will be developed based on the following systematic methodology:

- **Requirement Gathering & Analysis:** Conduct research on user preferences and industry best practices.
- **UI/UX Design:** Design user-friendly interfaces for web and mobile applications.

Development:

- **Frontend:** React.js for the website, Flutter for the mobile application.
- **Backend:** Node.js with Express.js.
- **Database:** MongoDB for structured data, Firebase for real-time use.
- **AI Chabot:** OpenAI API/Dialog flow integration.
- **Integration & Testing:** Seamless functioning and performance tuning.
- **Deployment & Maintenance:** Deploy the application and offer periodic updates.

5.Project Scope

The Project encompasses:

- **Website & Mobile App Development:** Feature-packed website and a mobile app with seamless synchronizing.
- **AI Chatbot Implementation:** Intelligent chatbot to aid users in cooking queries.
- **User Engagement Features:** Likes, comments, shares, and personal recipe uploads.

- **Video Tutorials:** Embedded video lessons for step-by-step cooking instructions.
- **Dynamic Recipe Database:** Admin-managed recipe addition and updates.
- **Real-Time Notifications:** New recipe and user activity alerts.
- **Scalability:** Potential expansion to add grocery lists, meal planning, and social community.

6. Brief Feasibility Study

The proposed system, consisting of a mobile application developed using Flutter and a web platform built on the MERN stack (React JS, Node.js, MongoDB, and Express), is both technically and economically feasible while being operational for end users. Leveraging modern technologies like Firebase for real-time data management and user attendance tracking ensures seamless performance, scalability, and responsiveness across devices. From a technical standpoint, the system is robust, user-friendly, and integrates well with cross-platform environments. Economically, the use of open-source frameworks significantly reduces development and maintenance costs, making the project financially viable. Operationally, the intuitive design and cross-device compatibility ensure ease of use for both administrators and end-users, enhancing productivity and ensuring smooth adoption without the need for extensive training.

7. Solution Application Areas

- **Home Cooks:** Find and follow recipes with ease using AI.
- **Health-Conscious Users:** Receive nutritional analysis and meal planning.
- **Cooking Schools:** Utilize as a teaching tool for students.
- **Food Bloggers & Influencers:** Share recipes and interact with the community.

8.Tools/Technology

- Website (MERN Stack) **Frontend:** React.js
- **Backend:** Node.js with Express.js
- **Database:** MongoDB
- **Authentication & Storage:** Firebase
- **AI Chatbot:** OpenAI API / Dialog flow
- Mobile Application (Flutter + Firebase)
- **Framework:** Flutter (Dart Language)
- **Backend:** Firebase Fire store (for real-time database)
- **Authentication:** Firebase Auth
- **Cloud Functions:** Firebase Functions (for AI chatbot processing)
- **Push Notifications:** Firebase Cloud Messaging (FCM)

9.Expertise of the Team

- **Frontend Developers:** Experienced in React.js and Flutter.
- **Backend Developers:** Skilled in Node.js, Express.js, and Firebase.
- **AI & ML Experts:** Chatbot development and NLP specialists.
- **Database Administrators:** MongoDB and Firebase Fire store experts.
- **Project Managers:** Ensuring smooth execution and on-time completion.

10. Milestone

Weeks	Description	Tools
Week 1-2	Requirements gathering and UI/UX design.	Figma
Week 3-5	Frontend development of website and API	Flutter, React
Week 6-8	Backend development and database configuration.	Node, firebase
Week 11-12	Implementation of AI Chabot and testing	AI Python
Week 13-14	Testing bug fixing	Testing

Reference

- OpenAI API documentation for integration of chatbot.
- Firebase documentation for authentication and real-time database.
- Mango dB official documentation.
- Flutter development guideline and best practices.
- Research Articles on AI-based recommendations system in the food sector

